Investigating Deep Q-Learning Modifications with Randomised Control Trials

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July 15, 2026



A proposal submitted to the Faculty of Science, University of the Witwatersrand, Johannesburg, in partial fulfilment of the requirements for the degree of Bachelor of Science with Honours

Abstract

Deep Reinforcement Learning (RL), specifically the Deep Q-Network architecture, has demonstrated tremendous success in mastering complex decision-making tasks through high-dimensional sensory input. Even though many algorithmic changes have been proposed to enhance the stability and sample efficiency of the original DQN, their individual contributions are frequently obscured by combinatorial implementations or inconsistent hyperparameter tuning across different studies. This research proposes a rigorous investigation into the isolated contribution of four established DQN variants – Prioritised Experience Replay (PER), Double DQN (DDQN), n-step learning, and Averaged DQN – using a Randomised Control Trial (RCT) methodology.

By defining a fixed control group based on the canonical Nature-DQN architecture and four independent treatment arms, this study aims to quantify the individual "treatment effect" of each modification on agent sample efficiency. The aforementioned efficiency will be measured as the Area Under the Curve (AUC) of the evaluation reward across the standardised **MinAtar** environment suite, specifically the games Breakout, Asterix, and Freeway. Statistical significance will be validated through paired samples t-tests across $N \geq 30$ independently sampled hyperparameter configurations to ensure internal validity against the stochastic nature of RL environments.

The primary contribution of this research is the provision of an isolated, statistically robust performance benchmark for DQN modifications, establishing a modular framework that facilitates the structured evaluation of future algorithmic enhancements.

Declaration

I, Matimu Khosa, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of Bachelor of Science with Honours in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Acknowledgements

I express my sincere gratitude to my supervisors, Steven James and Benjamin Rosman, for their invaluable guidance, patience, and continuous support throughout the formulation of this research proposal.

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List of Abbreviations

AUC	**A**rea **U**nder the **C**urve
DNN	**D**eep **N**eural **Network**
DRL	**D**eep **R**einforcement **L**earning
DQN	**D**eep **Q**-**N**etwork
RCT	**R**andomised **C**ontrol **T**rial
RL	**R**einforcement **L**earning
MDP	**M**arkovian **D**ecision **Process**
ML	**M**achine **L**earning

Chapter 1

Introduction

From autonomous driving systems that navigate complex urban traffic to robotic arms that master delicate manipulation tasks in manufacturing, the ability of machines to learn optimal decision-making strategies has profound implications for solving real-world problems. The foundation of these advancements lies in Reinforcement Learning (RL), a framework in which an artificial agent learns to make a sequence of decisions by interacting with an environment which is foreign to it. As formalised by Sutton *et al.* [1998], the agent learns through trial and error, receiving numerical reward signals that it seeks to maximise over time. Historically, RL struggled in environments with massive, complex state spaces, such as raw visual data. This limitation was overcome with the advent of Deep Reinforcement Learning (DRL). DRL handles high-dimensional inputs by combining traditional RL principles with deep artificial neural networks—powerful function approximators capable of extracting meaningful features directly from raw visual data. The seminal Deep Q-Network (DQN) introduced by Mnih *et al.* [2015] successfully bridged these fields, showing that a single algorithm could learn to play a variety of Atari 2600 games at a superhuman level solely by observing raw pixel inputs and game scores. Ever since this breakthrough, DRL has become the cornerstone of modern artificial intelligence, leading to its application in challenges such as optimising data centre cooling systems, advancing drug discovery, and managing complex financial portfolios.

Despite the widespread success and applicability of DRL, the field is currently battling with a severe methodological vulnerability: the reproducibility crisis [Henderson *et al.* 2018]. DRL algorithms, particularly those based on value-estimation like DQN, are exceptionally sensitive to stochastic factors. Minor changes in neural network weight initialisation, the random seed used by the environment simulator, or hyperparameter configurations can drastically change an agent's apparent performance. Currently, the DRL landscape is saturated with algorithmic modifications to the original Nature-DQN architecture—such as Double Deep Q-learning (DDQN) [Van Hasselt *et al.* 2016], Prioritised Experience Replay (PER) [Schaul *et al.* 2015], n-step learning [Hernandez-Garcia and Sutton 2019], and Averaged DQN [Anschel *et al.* 2017]. Each selected variant is frequently presented as achieving state-of-the-art results. However, previous literature has highlighted significant shortcomings in the way these claims are evalu-

ated. Notably, Machado *et al.* [2018] revisited the Arcade Learning Environment (ALE) evaluation protocols and found critical flaws in the literature: researchers frequently employ faulty metrics, allow deterministic starting states that agents can exploit by memorisation rather than generalisation, and report best-case single-seed runs rather than statistically robust metrics. Consequently, the individual contribution of any single algorithmic modification remains poorly understood. When a new DQN variant achieves a higher score, it is often impossible to determine whether the improvement is genuinely due to the architectural modification itself, or simply a byproduct of extensive, modification-specific hyperparameter tuning and favourable random seeds [Henderson *et al.* 2018; Pretorius *et al.* 2020].

To explicitly address the lack of rigorous standardisation and the statistical vulnerabilities identified by Machado *et al.* [2018], this research will implement a Randomised Control Trial (RCT) methodology to benchmark established DQN modifications. Adapted conceptually from clinical and pharmaceutical research to establish rigorous evaluation protocols in machine learning [Colas *et al.* 2018; Gorsane *et al.* 2022], an RCT framework is the gold standard for isolating causal relationships. In the context of this project, the RCT design is inspired by Pretorius *et al.* [2020], where the "individuals" of the trial are not single random seeds, but independently sampled hyperparameter configurations drawn from a shared population space. Each such configuration is evaluated under every algorithmic variant, and this experiment is repeated across $N \geq 30$ such individuals. This paired design directly controls for the confounding influence of hyperparameter choice, ensuring that any observed performance difference is attributable to the algorithm itself, rather than a favourable configuration. To evaluate the variance of these modifications in a computationally feasible and standardised manner, this study will utilise the MinAtar environment suite—a miniaturised derivative of the Arcade Learning Environment (ALE)—specifically focusing on the classic Atari games *Breakout*, *Asterix*, and *Freeway*. Rigorous statistical analysis, including paired samples t-tests and Bonferroni corrections, will be applied to the resulting AUC distributions [Kim 2015].

Through this framework, the project poses two research questions that aim to revolutionise the DRL field. Firstly, when evaluated under strict RCT conditions against a canonical Nature-DQN baseline—with hyperparameter configurations sampled from a shared population—do isolated algorithmic modifications yield a statistically significant improvement in sample efficiency? Instead of looking at maximum episodic reward, we will measure the Area Under the Curve (AUC) of the learning trajectory. Secondly, to what extent does controlling for hyperparameter optimisation and random initialisation via an RCT framework expose previously accepted modifications as statistically insignificant? By answering these questions, this research seeks to move DRL away from anecdotal performance claims and towards a rigorous, evidence-based science of algorithmic design.

1.1 Research Proposal Outline

The remainder of this proposal is structured as follows. Chapter 2 provides the foundational background of value-based DRL and reviews the architectural innovations that serve as our treatment arms. Chapter 3 formalises the problem statement, explicitly defining the research questions, hypotheses, and scope of this study. Chapter 4 details the quantitative Randomised Control Trial (RCT) methodology, including the experimental setup, environments, and statistical validation techniques. Chapter 5 outlines the proposed timeline for project execution and risk management strategies. Finally, Chapter 6 concludes the proposal by summarising the anticipated contributions of this research.

Chapter 2

Background and Related Work

The challenge of isolating genuine algorithmic progress in Deep Reinforcement Learning (DRL) from performance gains driven by hyperparameter tuning and stochasticity is far from solved. A common problem with proposed Deep Q-Network (DQN) variants is that they are frequently evaluated under inconsistent conditions, making it difficult to discern if a modification genuinely improves the learning process or merely exploits a specific random seed [Henderson *et al.* 2018]. Having access to a standardised, statistically rigorous benchmark would help researchers evaluate novel algorithms accurately and confidently.

The research question considered in this proposal is how to apply a Randomised Control Trial (RCT) methodology to benchmark established DQN modifications in isolation. A secondary question is how continuous, aggregate metrics—such as the Area Under the Curve (AUC) combined with parametric statistical tests—can replace unreliable point estimates to provide game-independent statistical validation.

Using the above questions as guiding concepts, we discuss the most applicable previous work on the topic of value-based Deep Reinforcement Learning. We attempt to show that the problem of isolating the true "treatment effect" of DQN modifications has not yet been adequately solved due to compounding architectural complexities. Finally, we attempt to show that how algorithms are currently evaluated and how results are reported leaves something to be desired in terms of objective, robust, and reproducible metrics.

2.1 Background

The theoretical underpinnings of Q-learning are rooted in Markov Decision Processes (MDPs) and the pursuit of the optimal action-value function [Bellman 1957]. At any discrete time step t, an agent perceives a state S_t, executes an action A_t, obtains a scalar reward R_{t+1}, and transitions to a new state S_{t+1}. This generates the fundamental transition tuple $(S_t, A_t, R_{t+1}, S_{t+1})$.

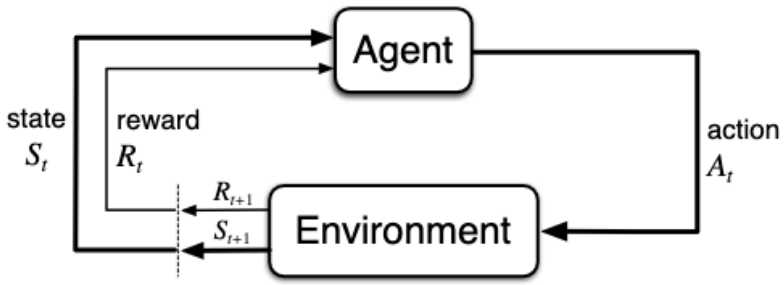


Figure 2.1: The standard Reinforcement Learning Agent-Environment interaction loop. The agent observes the state S_t, takes an action A_t, and receives a reward R_{t+1} alongside the subsequent state S_{t+1} [Sutton *et al.* 1998].

In tabular configurations, Q-learning iteratively updates action-value estimates via the Bellman Optimality Equation, eventually converging upon the optimal Q-function, $Q^*(s, a)$. The optimal action-value function is formally defined as:

$$Q^*(s, a) = \mathbb{E}\left[R_{t+1} + \gamma \max_{a'} Q^*(S_{t+1}, a')\right]$$

To thoroughly understand this foundational baseline, the independent components are defined as follows:

- $Q^*(s, a)$: The maximum expected return achievable from state s, taking action a, and following the optimal policy thereafter.
- $\mathbb{E}$: The expected value operator, accounting for the stochastic transitions of the environment.
- R_{t+1}: The immediate scalar reward received after taking action a.
- γ: The discount factor ($0 \leq \gamma \leq 1$), which dictates the present value of future rewards.
- $\max_{a'} Q^*(S_{t+1}, a')$: The maximum estimated value of the subsequent state S_{t+1}, representing the assumption that the agent will take the optimal action a' in the future.

While these tabular approaches provide robust convergence guarantees within discrete and finite environments, they succumb to the "curse of dimensionality" when applied to more expansive state spaces, such as the raw visual pixel arrays of video games [Sutton *et al.* 1998]. This constraint dictates the use of function approximators—a technique that, prior to the integration of deep neural networks, frequently resulted in learning instability and divergence.

2.2 The Canonical Deep Q-Network (Control Group)

Comprehending the foundational Deep Q-Network (DQN) [Mnih *et al.* 2015] requires understanding how it stabilised non-linear function approximation within high-dimensional spaces. This seminal work identified major reinforcement learning obstacles, including data non-stationarity and highly correlated observation sequences, and mitigated them using two core components. These components form the immutable algorithmic baseline of our RCT:

1. **Experience Replay:** This technique disrupts temporal correlations in the observed data by archiving transition tuples in a dedicated memory buffer, $\mathcal{D}$. Throughout the training process, the algorithm uniformly samples mini-batches from this buffer, denoted as $U(\mathcal{D})$. This procedure not only mitigates the variance of updates but also bolsters sample efficiency.
2. **Fixed Target Networks:** To prevent moving learning targets, the architecture incorporates a separate target network with independent parameters θ^-. These weights are periodically copied from the active online network parameters θ, but are otherwise kept frozen during intermediate updates. This strategy significantly suppresses learning oscillations.

Integrating these elements, the objective function minimised at training iteration i is formulated as:

$$L_i(\theta_i) = \mathbb{E}_{(s,a,r,s')\sim U(\mathcal{D})}\left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i)\right)^2\right]$$

The independent components of this loss function are:

- $L_i(\theta_i)$: The loss to be minimised at training iteration i.
- $\mathbb{E}_{(s,a,r,s')\sim U(\mathcal{D})}$: The expectation over a mini-batch of transitions sampled uniformly U from the replay buffer $\mathcal{D}$.
- $r+\gamma \max_{a'} Q(s', a'; \theta_i^-)$: The Temporal Difference (TD) target, acting as the "ground truth" for the update, calculated using the frozen target network weights θ_i^-.
- $Q(s, a; \theta_i)$: The current prediction of the action-value using the active online network weights θ_i.

This canonical formulation acts as a highly stable, universally understood benchmark against which all treatment arms will be measured.

2.3 Broader Architectural Innovations in DRL

To address the inherent flaws of the baseline DQN, the research community has proposed numerous algorithmic improvements. Moving beyond data sampling and target adjustments, several prominent innovations sought to reconstruct the internal neural architecture entirely.

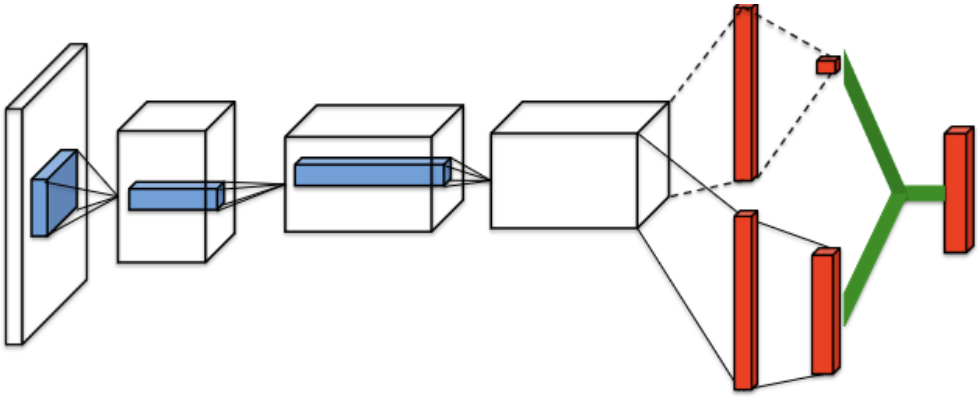


Figure 2.2: The Dueling Network Architecture. The convolutional layers process the spatial state, after which the fully connected layers bifurcate into two separate streams: one for estimating the state-value function $V(s)$, and another for the advantage function $A(s, a)$ [Wang *et al.* 2016].

Dueling DQN implements a structural bifurcation that separates the estimation of state values from the advantages of individual actions [Wang *et al.* 2016]. Dividing the network into two specialised streams allows the algorithm to recognise inherently valuable states regardless of the action taken (Figure 2.2). A more radical theoretical departure involved transitioning from expected scalar returns to comprehensive probability distributions. Distributional RL alters this fundamental approach by replacing the scalar Q-value with a full value distribution [Bellemare *et al.* 2017]. Expanding on this paradigm, Implicit Quantile Networks (IQN) remove the limitations associated with fixed categorical bins, leveraging quantile regression to estimate the continuous quantile function of the target return [Dabney *et al.* 2018].

Furthermore, effective environmental exploration has been addressed by Noisy Networks (NoisyNets), which abandon the traditional ϵ-greedy exploratory approach by injecting parameterised Gaussian noise straight into the network's weights [Fortunato *et al.* 2017]. To handle partially observable environments, Recurrent Replay Distributed DQN (R2D2) introduces Long Short-Term Memory (LSTM) layers to the architecture [Kapturowski *et al.* 2019].

Acknowledging that these diverse modifications address largely independent challenges, researchers integrated several of them into a unified framework [Hessel *et al.* 2018]. This resulting "Rainbow" model illustrated that these algorithmic enhancements are highly complementary, setting new benchmarks for data efficiency on standard Atari environments.

However, for an RCT aiming to isolate algorithmic utility without confounding variables, these broader innovations must be excluded. Modifying the neural topology (as in Dueling or R2D2) or altering the fundamental loss function (as in Distributional RL) deviates too far from the canonical Nature-DQN baseline. Evaluating them would obscure the isolated effects of simpler algorithmic logic.

2.4 Algorithmic Extensions as Treatment Arms

Consequently, this project selects modular modifications that maintain the strict Nature-DQN neural topology. These variants represent the independent variables—the treatment arms—whose isolated efficacy must be quantified. The subsections below introduce the mathematical mechanism of each selected variant. Their formal placement within the RCT experimental design—including how they are applied to each sampled individual—is detailed in Chapter 4.

2.4.1 Double DQN

The standard DQN frequently suffers from an overestimation bias because the target computation relies on the same network weights to both select and evaluate the greedy action. Double DQN (DDQN) mitigates this positive bias by decoupling the action selection mechanism from its subsequent evaluation [Van Hasselt *et al.* 2016]. Specifically, the online network identifies the most promising action, while the frozen target network assesses its value:

$$Y_t^{DDQN} = R_{t+1} + \gamma Q(S_{t+1}, \arg\max_a Q(S_{t+1}, a; \theta); \theta^-)$$

The independent components of this adjustment are:

- Y_t^{DDQN}: The revised target value for Double DQN.
- $\arg\max_a Q(S_{t+1}, a; \theta)$: The online network ($\theta$) selects the action a that it currently believes is optimal for the next state.
- $Q(\ldots; \theta^-)$: The target network (θ^-) evaluates the specific action chosen by the online network, rather than simply taking the maximum possible value.

2.4.2 Prioritised Experience Replay (PER)

While DDQN refines the target value itself, Prioritised Experience Replay (PER) optimises the data sampling process used to achieve that target. Rather than extracting experiences uniformly, PER biases the sampling probability according to the magnitude of a transition's temporal-difference (TD) error, δ_i. The likelihood of selecting a specific transition i is defined by [Schaul *et al.* 2015]:

$$P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha}$$

The independent components dictating this sampling are:

- $P(i)$: The sampling probability of transition i.
- p_i: The priority of transition i, defined as $p_i = |\delta_i| + \epsilon$, where $|\delta_i|$ is the absolute TD-error and ϵ is a small constant ensuring all transitions have a non-zero probability.

- α: A hyperparameter determining how much prioritisation is used ($\alpha = 0$ corresponds to the uniform sampling of the control group).
- $\sum_k p_k^\alpha$: The normalising denominator summing the priorities of all transitions currently in the replay buffer.

2.4.3 n-step Learning

Moving beyond 1-step targets, the temporal scope of the learning update can be broadened using n-step returns [Hernandez-Garcia and Sutton 2019]. By accumulating actual empirical rewards over a horizon of n steps before bootstrapping, the algorithm accelerates the backward flow of rewards:

$$G_{t:t+n} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n \max_a Q(S_{t+n}, a; \theta^-)$$

The independent components of the n-step target are:

- $G_{t:t+n}$: The formulated n-step return target.
- $\sum_{k=0}^{n-1} \gamma^k R_{t+k+1}$: The accumulation of actual, discounted empirical rewards observed over exactly n steps into the future.
- $\gamma^n \max_a Q(S_{t+n}, a; \theta^-)$: The bootstrapped value estimation from the target network, applied only at the n-th state rather than the immediate next state.

2.4.4 Averaged DQN

To directly combat target instability and approximation variance, Averaged DQN alters the target calculation by maintaining an ensemble of previous networks [Anschel *et al.* 2017]. Instead of relying on a single, periodically updated target network, Averaged DQN calculates the target as the mean maximum Q-value across K previously stored target networks (θ_k^-):

$$Y_t^{Ave} = R_{t+1} + \gamma \frac{1}{K} \sum_{k=1}^{K} \max_{a'} Q(S_{t+1}, a'; \theta_k^-)$$

The independent components of this ensemble approach are:

- Y_t^{Ave}: The smoothed target value for Averaged DQN.
- K: The total number of historical target networks stored in the ensemble.
- $\frac{1}{K} \sum_{k=1}^{K}$: The arithmetic mean operator, averaging the Q-value estimates.
- θ_k^-: The parameters of the k-th previously saved target network.

2.5 Evaluation and the Reproducibility Crisis

The training and evaluation signals used to benchmark these methods are critically important. Current appraisals of reinforcement learning literature point out major shortcomings in standard evaluation methodologies [Henderson *et al.* 2018].

2.5.1 Methodological Vulnerabilities and the RCT Framework

Henderson *et al.* [2018] outlined a severe crisis of reproducibility, noting that reported performance gains are often artefacts of specific random seed choices and extensive hyper-parameter optimisation, rather than fundamental algorithmic superiority. This was further supported by Machado *et al.* [2018], who revisited the Arcade Learning Environment (ALE) evaluation protocols. They found that researchers frequently employ inconsistent evaluation metrics and allow deterministic starting states that agents can exploit by memorisation.

The Randomised Control Trial (RCT) framework serves as the direct methodological antidote to these vulnerabilities. Rather than fixing a single "best" hyperparameter configuration and reporting the resulting score, an RCT treats the space of possible hyperparameter settings as a *population* from which individuals can be sampled. This insight, drawn from the RCT approach applied to deep learning by Pretorius *et al.* [2020], means that conclusions about an algorithm's efficacy hold across the broader population of plausible configurations, rather than being contingent on a single, potentially cherry-picked setting. The full experimental instantiation of this framework is detailed in Chapter 4.

2.5.2 Environmental Standardisation via MinAtar

To practically execute this RCT and resolve the computational constraints associated with standard ALE evaluations, this research leverages the MinAtar environment suite. MinAtar provides miniaturised versions of classic Atari games, abstracting the observation space into a $10 \times 10 \times C$ Boolean array. This removes the prohibitive computational overhead of processing raw, high-resolution pixels through deep convolutional layers while retaining the complex spatial and temporal dynamics of the original games.

Crucially, this computational efficiency is what makes running $N \geq 30$ independent trials feasible within the constraints of this research. While a limitation of MinAtar is that it simplifies the representation learning challenge—meaning results may not scale identically to visually complex 3D environments—it perfectly isolates the value-estimation and decision-making mechanics required to accurately benchmark DQN modifications.

2.5.3 Final Evaluation and Statistical Validation

Judging an algorithm based on point estimates derived from a minimal set of random seeds creates a statistically distorted view of its true efficacy [Agarwal *et al.* 2021]. To ensure rigorous validation, evaluations must transition to aggregate metrics over large

sample sizes. Measuring the Area Under the Curve (AUC) of the learning trajectory captures sample efficiency over the entire training lifecycle.

Furthermore, because all algorithmic variants are applied to each sampled individual (hyperparameter configuration), the resulting AUC values are naturally paired. This enables the application of paired statistical tests, which provide greater statistical power by controlling for the individual-level variability introduced by different hyperparameter configurations. The precise statistical procedures are described in Chapter 4.

Chapter 3

Research Questions and Aims

This chapter builds upon the methodological vulnerabilities discussed in Chapter 2 by detailing the current evaluation landscape, describing the ideal standard for algorithmic benchmarking, and explicitly outlining how this research aims to bridge that gap. Finally, we specify our overarching research questions, hypotheses, actionable objectives, and the inherent limitations of the proposed study.

3.1 Problem Formulation: The Current vs. Ideal Situation

In an ideal research environment, Artificial Intelligence researchers would be able to propose novel Deep Reinforcement Learning (DRL) algorithms and confidently evaluate them against a universally standardised baseline. This ideal framework would reliably separate genuine architectural utility from the "noise" of hyperparameter tuning, allowing for the direct, empirical comparison of algorithmic modifications. Furthermore, this framework would utilise objective, game-independent metrics to rigorously evaluate whether a new variant constitutes a statistically significant advancement.

Currently, the field operates far from this ideal. As established in Chapter 2, the DRL landscape is hindered by a severe reproducibility crisis. Evaluating a new Deep Q-Network (DQN) variant typically involves combinatorial hyperparameter tuning specific to that variant, testing it on a handful of arbitrary random seeds, and reporting the maximum episodic reward. Consequently, when an algorithm like Double DQN or Prioritised Experience Replay (PER) is integrated into a system and achieves a high score, it is often impossible to determine whether the success is due to the mathematical modification itself or the extensive, variant-specific tuning that accompanied it [Henderson *et al.* 2018].

To move toward the ideal situation, this research proposes the application of a Randomised Control Trial (RCT) methodology. By sampling hyperparameter configurations from a shared population space and applying all algorithmic variants to each sampled

configuration, this framework strips away the confounding influence of hyperparameter choice.

3.2 Research Questions

To systematically address the lack of rigorous standardisation in value-based DRL, this research poses two primary questions:

1. **Question 1 (Algorithmic Efficacy):** When evaluated under strict RCT conditions against a canonical Nature-DQN baseline—with hyperparameter configurations drawn from a shared population—do isolated, topology-preserving modifications—specifically Double DQN, Prioritised Experience Replay, n-step learning, and Averaged DQN—yield a statistically significant improvement in sample efficiency?

2. **Question 2 (Methodological Validity):** Can the utilisation of continuous aggregate metrics (Area Under the Curve) combined with parametric statistics (paired samples t-tests across sampled hyperparameter populations) effectively expose the variance and unreliability of traditional point-estimate evaluations in DRL?

3.3 Hypotheses

We hypothesise that implementing an RCT methodology will significantly alter the perceived efficacy of established DQN variants. Specifically, we predict that the isolated "treatment effects" of these modifications will yield substantially smaller margins of improvement compared to their combinatorially tuned counterparts frequently reported in the literature.

Secondly, we hypothesise that measuring the Area Under the Curve (AUC) across a diverse population of hyperparameter configurations ($N \geq 30$ sampled individuals) will provide a fundamentally more robust metric than maximum episodic reward. We expect this statistical framework to successfully identify algorithms whose occasional high performance is merely a byproduct of a favourable hyperparameter configuration, filtering out such statistical noise from genuine algorithmic progress.

3.4 Research Aims and Objectives

3.4.1 Research Aims

The aims of this research are twofold. Firstly, it aims to construct and execute a rigorous Randomised Control Trial framework capable of benchmarking value-based DRL algorithms across a population of hyperparameter configurations. Secondly, it aims to demonstrate the value of continuous aggregate metrics and parametric statistical validation over unreliable point estimates.

3.4.2 Objectives

To achieve these aims, the project requires the execution of the following concrete subtasks:

- Define a shared hyperparameter population space from which experimental individuals are sampled.
- Implement a universally reproducible Control Group adhering strictly to the canonical 2015 Nature-DQN architecture.
- Implement four distinct Treatment Arms (Double DQN, PER, n-step learning, and Averaged DQN), ensuring they seamlessly interface with the Control Group's neural topology and can be applied to any sampled individual.
- Integrate the MinAtar environment suite (*Breakout, Asterix, Freeway*) to standardise the observation space and ensure computationally feasible high-volume testing.
- Execute all algorithmic variants for each of the $N \geq 30$ sampled individuals.
- Calculate the Area Under the Curve (AUC) for every run.
- Validate the resulting data by performing paired samples t-tests, comparing the AUC of each Treatment Arm directly against the Control Group on a per-individual basis, and apply Bonferroni corrections to control for multiple comparisons.

3.5 Scope and Limitations

Certain constraints define the boundaries of this study. Firstly, to ensure the RCT remains fundamentally sound, we are limited to evaluating modular algorithmic variants that do not require overhauling the underlying neural network architecture. Consequently, broader structural innovations such as Dueling Networks or Distributional RL fall outside the scope of this specific trial.

Secondly, the standardisation and computational feasibility required to execute $N \geq 30$ sampled individuals dictate the use of the MinAtar environment suite. While MinAtar perfectly isolates the value-estimation and decision-making mechanics of Q-learning, its simplified $10 \times 10 \times C$ Boolean observation space reduces the burden of representation learning. Therefore, the magnitude of the treatment effects observed in this study may not scale identically when applied to visually complex, high-resolution 3D environments.

Finally, while an RCT framework aims to eliminate all external variables, neural network training inherently contains uncontrollable stochasticity at the hardware level (e.g., GPU floating-point non-determinism). We attempt to mitigate this by averaging over a large sample size, but it remains an acknowledged limitation of deep learning benchmarking.

Chapter 4

Research Methodology

This chapter details the specific experimental procedures, environments, and statistical methods required to execute the Randomised Control Trial (RCT) outlined in the previous chapters. It provides a comprehensive breakdown of the research design, ensuring the benchmarking of the Deep Q-Network (DQN) modifications is transparent, rigorous, and fully reproducible.

4.1 Research Design

This project follows a quantitative, empirical research design structured as a Randomised Control Trial, drawing conceptual inspiration from the paired RCT approach applied to deep learning by Pretorius *et al.* [2020]. The central insight of this design is that evaluating algorithms under a *single* fixed hyperparameter configuration conflates algorithmic merit with configuration-specific luck. Instead, we treat the space of plausible hyperparameter settings as a **population**, and each randomly drawn configuration as an **individual** within that population. The question we ask is not "does DDQN beat DQN at this one configuration?" but rather "across the realistic population of configurations a practitioner might use, does DDQN systematically outperform DQN?"

4.1.1 Population: The Hyperparameter Space

The population is defined as the Cartesian product of discrete candidate values for each shared training hyperparameter. Because all algorithmic variants are built on the canonical Nature-DQN topology, these hyperparameters govern the shared aspects of training: network capacity, optimisation, and experience collection. The hyperparameter space is defined in Table 4.1.

An **individual** is a single configuration sampled uniformly at random from this space. Importantly, some individuals will represent poor configurations (e.g., an excessively high learning rate combined with a small replay buffer), while others will represent strong ones. This mirrors the clinical analogy: trial participants vary in their baseline

Hyperparameter	Candidate Values
Learning rate α	$\{0.0001,\ 0.0005,\ 0.001\}$
Replay buffer size $\lvert\mathcal{D}\rvert$	$\{10\,000,\ 50\,000,\ 100\,000\}$
Mini-batch size	$\{32,\ 64\}$
Target network update frequency (steps)	$\{500,\ 1\,000,\ 2\,000\}$
ϵ-decay duration (steps)	$\{50\,000,\ 100\,000,\ 200\,000\}$
Number of hidden layers	$\{1,\ 2\}$

Table 4.1: The shared hyperparameter population space. An individual is a single configuration sampled uniformly at random from this space.

characteristics, but the randomisation ensures that, on average, every treatment arm encounters the same distribution of individuals.

4.1.2 Treatment Arms and Control Group

For each sampled individual, *all five* algorithmic variants are trained under that exact hyperparameter configuration. This paired within-subjects design is the critical mechanism that controls for hyperparameter confounding. Table 4.2 summarises the control group and the four treatment arms evaluated in this study.

Algorithm	Modification Target	Primary Mechanism
Baseline DQN	None (Control)	Uniform sampling, 1-step target.
Double DQN	Target Calculation	Decouples action selection from evaluation.
PER	Data Sampling	Biases sampling by TD-error magnitude.
n-step DQN	Temporal Horizon	Accumulates n rewards before bootstrapping.
Averaged DQN	Target Calculation	Averages targets across K historical networks.

Table 4.2: A comparison of the canonical baseline against the selected RCT treatment arms. All five variants are applied to every sampled individual.

The experiment is then repeated for $N \geq 30$ independently sampled individuals, ensuring that the results are robust to the variance of the hyperparameter population. Figure 4.1 provides a visual overview of this paired RCT structure.

4.2 Instruments and Environments

4.2.1 The MinAtar Environment Suite

To evaluate the agents, this study will utilise the MinAtar environment suite, specifically focusing on the classic arcade games *Breakout*, *Asterix*, and *Freeway*. MinAtar is explicitly chosen over the standard Arcade Learning Environment (ALE) because it reduces the observation space to a $10 \times 10 \times C$ Boolean array. This massively reduces the

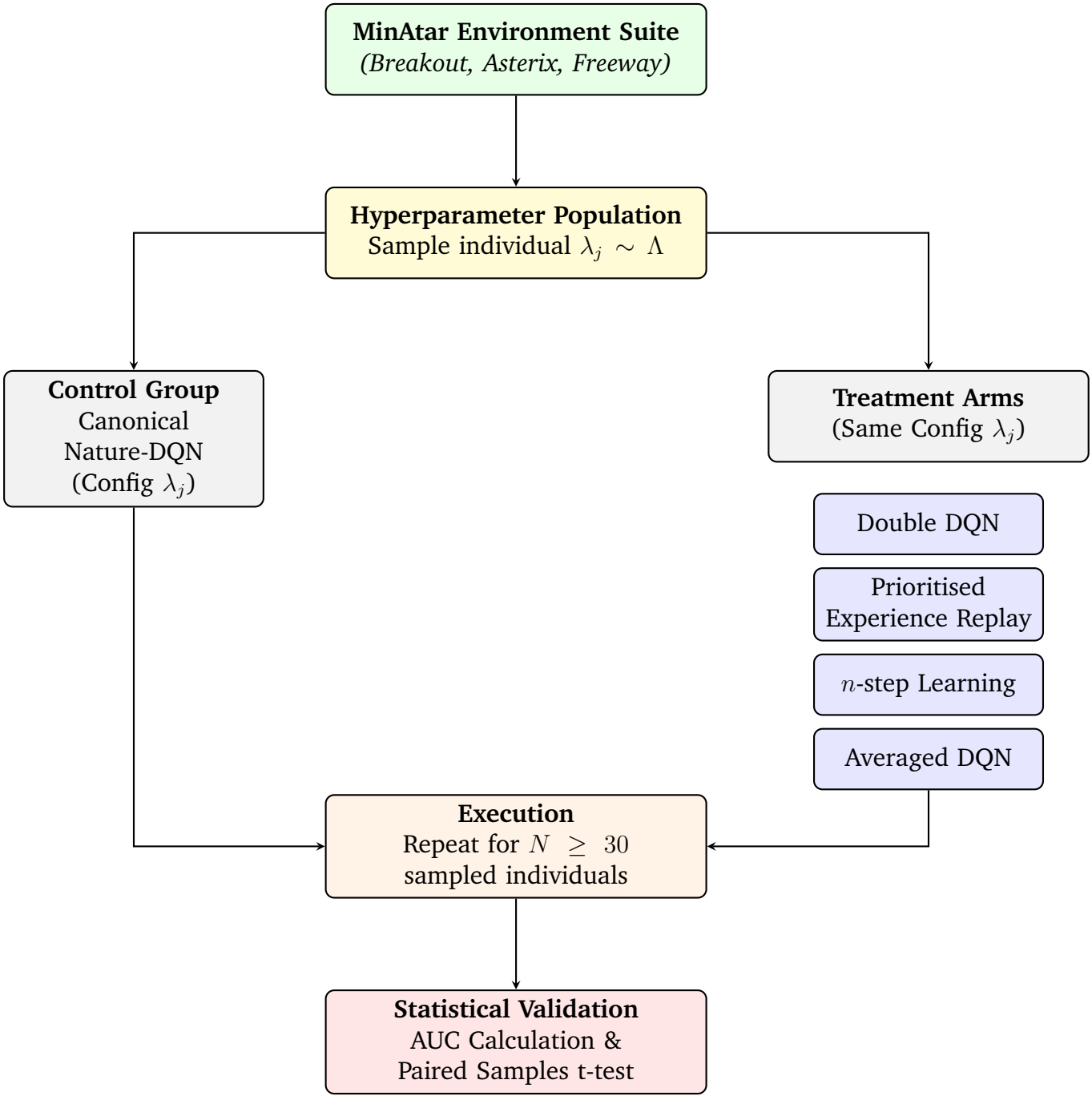


Figure 4.1: A visual representation of the proposed Randomised Control Trial (RCT) framework. An individual is a hyperparameter configuration sampled from the shared population Λ. All five algorithmic variants are trained under the same configuration, and this process is repeated for $N \geq 30$ individuals. Statistical validation is performed on the paired AUC distributions.

computational overhead associated with deep convolutional feature extraction while preserving the complex spatial and temporal dynamics required to test value-based decision making.

This specific trio of environments was deliberately selected to confront the DRL agents with a diverse set of Markov Decision Process (MDP) challenges. This ensures that an algorithmic modification is robust across different styles of gameplay, rather than merely over-indexing on a specific type of environment:

- ***Breakout*:** Requires precise timing and spatial trajectory prediction. Because the agent receives immediate feedback for breaking a block, it serves as an excellent benchmark for *dense reward* processing and continuous control.
- ***Asterix*:** Demands rapid reactive navigation and immediate spatial awareness to collect items while dodging continuously spawning enemies. This tests an agent's ability to learn highly dynamic, fast-twitch policies under temporal pressure.
- ***Freeway*:** Presents a significant challenge in patience and delayed gratification. The agent must navigate across lanes of moving traffic but only receives a reward upon successfully reaching the opposite side. This serves as a vital benchmark for *sparse reward* learning and long-horizon planning.

By evaluating the Treatment Arms across this specific spectrum—from dense to sparse rewards, and from reactive dodging to predictive planning—we can confidently assess the true generalisability of the proposed algorithmic improvements.

4.3 Algorithmic Setup and Data Collection

Each algorithmic variant will be trained for a standardised duration of 5×10^6 environmental frames under each sampled individual configuration. During training, the agent's performance will be periodically evaluated every 5×10^4 frames by freezing the network weights and running 10 independent, strictly greedy evaluation episodes (where the exploration rate $\epsilon = 0$).

To ensure that the observed performance variance stems entirely from the algorithms and hyperparameter configurations rather than environmental noise, these 10 evaluation episodes will be initialised using a fixed set of deterministic environmental seeds. The average episodic return of these 10 fixed-seed episodes will be recorded as a single data point on the agent's learning trajectory.

4.4 Statistical Analysis

4.4.1 Area Under the Curve (AUC)

To transition away from unreliable point estimates—such as reporting an algorithm's single highest evaluation score—this research will measure the Area Under the Curve (AUC) of the resulting learning trajectory. By taking the integral of the performance

curve over the entire training period, the AUC acts as a comprehensive proxy for *sample efficiency*. An agent that learns faster and maintains stability will yield a significantly higher AUC than an agent that learns slowly or suffers from catastrophic policy collapse.

4.4.2 Paired Samples t-tests and Bonferroni Corrections

Upon completion of training for all $N \geq 30$ sampled individuals, we possess a *paired* collection of AUC values: for each individual j, we have an AUC for the Control Group and an AUC for each Treatment Arm, all produced under the same hyperparameter configuration λ_j. This pairing is the direct statistical benefit of the RCT design—it allows us to compute a per-individual difference score $d_j = \text{AUC}^{(j)}_{\text{treatment}} - \text{AUC}^{(j)}_{\text{control}}$ and test whether the mean of these differences is significantly different from zero.

To do so, we will employ a **paired samples t-test** [Kim 2015]. The use of this parametric test is mathematically justified by the Central Limit Theorem, as our sample size of $N \geq 30$ individuals allows the sampling distribution of the mean difference to approximate a normal distribution. Prior to executing the t-tests, a Shapiro-Wilk test will be conducted on the difference scores to empirically verify this normality assumption.

We define our initial threshold for statistical significance at $\alpha = 0.05$. However, because we are testing four distinct Treatment Arms against a single Control Group, the probability of encountering a false positive (Type I error) inflates. To correct for this multiple-comparison problem, we will apply a **Bonferroni Correction**, adjusting our significance threshold to $\alpha_{\text{adjusted}} = \frac{0.05}{4} = 0.0125$. Therefore, an algorithmic modification will only be deemed a statistically significant advancement if its paired t-test yields a p-value ≤ 0.0125.

Chapter 5

Schedule of Work

This chapter outlines the proposed timeline for the execution of this research project. The schedule is structured to accommodate the rigorous development, mass computational execution, and statistical analysis required by the Randomised Control Trial (RCT) methodology. Additionally, we identify potential risks and outline mitigation strategies to ensure the project remains on track for the final submission.

5.1 Proposed Timeline

The research project will commence on Tuesday, 22 July, and conclude on Tuesday, 28 October. The timeline is divided into distinct, actionable phases: Baseline Implementation, Treatment Development, Mass Execution, Statistical Analysis, and Documentation.

Due to the heavy computational burden of executing $N \geq 30$ sampled individuals across multiple algorithms and environments, the mass execution phase has been allocated the largest segment of time.

5.2 Risk Management and Potential Issues

Empirical deep learning research is susceptible to a range of unpredictable delays and failure modes. The following risks have been identified, assessed for their potential impact, and paired with concrete mitigation strategies.

- **Cluster Downtime or Queue Delays:** The Wits computing cluster is a shared resource. High traffic or maintenance downtime could stall the mass execution phase and compress the time available for analysis and writing. *Mitigation:* The 4-week execution window provides a substantial buffer. Jobs will be scheduled incrementally rather than in a single massive batch, ensuring partial data is continually generated and that early results can inform debugging.
- **Algorithmic Instability or Silent Bugs:** A subtle implementation error in a treatment arm—for example, an incorrect priority update rule in PER or an off-by-one

Project Phase & Objectives	Start Date	End Date	Duration
Phase 1: Environment & Baseline Setup • Integrate the MinAtar suite (*Breakout, Asterix, Freeway*). • Implement and verify the Control Group. • Define and validate the hyperparameter population space.	22 July	05 August	2 weeks
Phase 2: Treatment Arm Implementation • Implement Double DQN, PER, n-step, and Averaged DQN. • Conduct local small-scale debugging runs.	06 August	26 August	3 weeks
Phase 3: Mass Execution (Cluster Computations) • Deploy the training runs to the Wits cluster. • Monitor training stability and log evaluation metrics.	27 August	23 September	4 weeks
Phase 4: Data Aggregation & Statistical Analysis • Extract and format learning trajectory data. • Calculate Area Under the Curve (AUC) for all runs. • Execute Shapiro-Wilk and paired samples t-tests with Bonferroni correction.	24 September	07 October	2 weeks
Phase 5: Results Synthesis & Drafting • Graph the learning curves and statistical distributions. • Draft the Results, Discussion, and Conclusion chapters.	08 October	21 October	2 weeks
Phase 6: Final Review & Submission • Conduct final proofreading and reference checks. • Submit the finalised research report and prepare poster.	22 October	28 October	1 week

Table 5.1: Proposed 14-week schedule of work for the completion of the research project.

error in the n-step return buffer—could invalidate weeks of cluster training without producing any obvious crash or error message. This is arguably the most serious risk to the internal validity of the study. *Mitigation:* Phase 2 includes mandatory small-scale debugging runs on a subset of individual configurations. Each treatment arm must demonstrate clear learning convergence and produce Q-value estimates consistent with theoretical expectations before being pushed to the cluster. Additionally, unit tests will be written for each algorithmic modification in isolation.

- **Insufficient Hyperparameter Population Coverage:** If the sampled individuals happen to cluster in a narrow, unrepresentative region of the hyperparameter space—for example, all samples drawing a high learning rate—the conclusions of the RCT may not generalise to the full population. *Mitigation:* A stratified sampling procedure will be adopted where possible, ensuring that the marginal distribution of each hyperparameter in the sampled set approximates the uniform distribution over its candidate values. The final sample will be audited before cluster submission.

- **Violations of Statistical Assumptions:** The paired t-test relies on the difference scores being approximately normally distributed. If the AUC distributions are heavily skewed or bimodal—a plausible outcome if some hyperparameter configurations produce catastrophic failure—this assumption may be violated. *Mitigation:* A Shapiro-Wilk normality test will be applied to the difference scores prior to each t-test. Should normality be rejected, non-parametric alternatives such as the Wilcoxon signed-rank test will be used in its place, preserving the paired structure of the design.

- **Computational Budget Overrun:** The number of training runs scales with the number of algorithms, environments, and sampled individuals. If the compute time per run is underestimated, it may not be possible to complete all $N \geq 30$ individuals within the 4-week execution window. *Mitigation:* A pilot timing experiment will be conducted in Phase 1 on a single individual and environment to obtain an accurate per-run runtime estimate. If the total budget is at risk, the number of environments will be reduced from three to two (prioritising *Freeway* and *Breakout* as the sparse and dense reward representatives), rather than reducing N below the statistically justified threshold.

- **Data Storage and Memory Constraints:** Saving network checkpoints and full replay buffers for a large number of agents can rapidly exhaust available storage quotas on the cluster. *Mitigation:* Only the scalar evaluation metrics required for the AUC calculation will be saved to disk. Network checkpoints will be discarded unless a run exhibits an anomalous learning curve that requires post-hoc debugging.

Chapter 6

Conclusion

Since the inception of the foundational Deep Q-Network (DQN), the architectures underpinning Deep Reinforcement Learning (DRL) have advanced considerably. However, as highlighted throughout this proposal, the rapid proliferation of algorithmic modifications has coincided with a pervasive reproducibility crisis. The widespread reliance on inconsistent testing protocols, combinatorial hyperparameter tuning, and unreliable point estimates has frequently masked genuine algorithmic strides, complicating the distinction between true innovation and statistical noise.

To rectify these methodological shortcomings, this research champions the application of a stringent Randomised Control Trial (RCT) methodology to benchmark DRL algorithms. Critically, this framework moves beyond simply fixing a single "best" hyperparameter configuration. Instead it treats the space of plausible training configurations as a population and samples individuals from it, applying all algorithmic variants—Double DQN, Prioritised Experience Replay, n-step learning, and Averaged DQN—to each individual. This paired design directly controls for the confounding influence of hyperparameter choice, ensuring that any observed performance difference is attributable to the algorithm's mechanism rather than a favourable configuration. The integration of the computationally efficient MinAtar environment suite makes executing this design across $N \geq 30$ individuals feasible.

Beyond the execution of the algorithms themselves, this research addresses a secondary, critical gap in the literature: how algorithms are empirically evaluated. By transitioning away from maximum episodic reward and instead measuring the Area Under the Curve (AUC) of the learning trajectory, this study establishes a robust proxy for sample efficiency. Validating these paired AUC distributions through paired samples t-tests with Bonferroni corrections ensures that any recorded performance improvements are statistically significant, not merely artefacts of favourable hyperparameter configurations or environmental initialisation.

Ultimately, translating the rigorous protocols of clinical RCTs into the domain of machine learning will yield unbiased, empirically sound evaluations of targeted DQN modifications. By shifting the evaluation paradigm away from anecdotal performance

claims, this research seeks to advance the broader objective of cementing reliable, evidence-based testing standards across the field of Deep Reinforcement Learning.

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